13

Evaporation and distillation

In common with other unit operations that are intended to separate components of foods (Chapter 6), evaporation and distillation aim to separate specific components to increase the value of the food. In both types of operation, separation is achieved by exploiting differences in the vapour pressure (volatility) of the components and using heat to remove one or more from the bulk of the food.

13.1 Evaporation

Evaporation, or concentration by boiling, is the partial removal of water from liquid foods by boiling off water vapour. It increases the solids content of a food and hence preserves it by a reduction in water activity (Chapter 1). Evaporation is used to pre-concentrate foods (for example fruit juice, milk and coffee) prior to drying, freezing or sterilisation and hence to reduce their weight and volume. This saves energy in subsequent operations and reduces storage, transport and distribution costs. There is also greater convenience for the consumer (for example fruit drinks for dilution, concentrated soups, tomato or garlic pastes, sugar) or for the manufacturer (for example liquid pectin, fruit concentrates for use in ice cream or baked goods). Changes to food quality that result from the relatively severe heat treatment are minimised by the design and operation of the equipment. Evaporation is more expensive in energy consumption than other methods of concentration (membrane concentration (Chapter 6)) and freeze concentration (Chapter 22) but a higher degree of concentration can be achieved (Table 13.1).

13.1.1 Theory

During evaporation, sensible heat is transferred from steam to the food, to raise the temperature to its boiling point. Latent heat of vaporisation is then supplied by the steam to form bubbles of vapour, which leave the surface of the boiling liquid. The rate of evaporation is determined by both the rate of heat transfer into the food and the rate of

Table 13.1 A comparison of energy efficiency and degree of concentration in different methods of concentration

	Steam equivalent (cost per kilogram of water removed divided by equivalent cost of steam)	Maximum concentration possible (%)
Ultrafiltration	0.001	28
Reverse osmosis	0.028	30
Freeze concentration	0.090–0.386	40
Evaporation		
Triple effect without aroma recovery	0.370	80
Triple effect with aroma recovery	0.510	80

From Thijssen (1974).

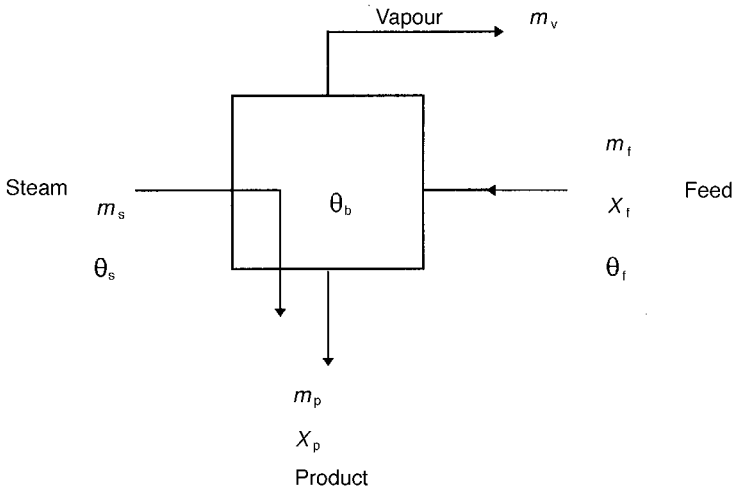


Fig. 13.1 Steady state operation of an evaporator: m_f (kg s^{-1}), mass transfer rate of feed liquor; m_p (kg s^{-1}), mass transfer rate of product; X_f, solids fraction of feed liquor; X_p, solids fraction of feed product; m_v (kg s^{-1}), mass transfer rate of vapour produced; m_s (kg s^{-1}), mass transfer rate of steam used; θ_f (ºC), initial feed temperature; θ_b (ºC), boiling temperature of food; θ_s (ºC), temperature of steam.

mass transfer of vapour from the food. These processes are represented schematically in Fig. 13.1.

Heat and mass balances

Heat and mass balances (Chapter 1) are used to calculate the degree of concentration, energy use and processing times in an evaporator. The mass balance states that 'the mass of feed entering the evaporator equals the mass of product and vapour removed from the evaporator'. For the water component, this is given by:

$$m_f(1 - X_f) = m_p(1 - X_p) + m_v \qquad 13.1$$

For solutes, the mass of solids entering the evaporator equals the mass of solids leaving the evaporator:

$$m_f X_f = m_p X_p \qquad 13.2$$

The total mass balance is $m_f = m_p + m_v$ 13.3

Assuming that there are negligible heat losses from the evaporator, the heat balance states that 'the amount of heat given up by the condensing steam equals the amount of heat used to raise the feed temperature to boiling point and then to boil off the vapour':

$$\begin{aligned} Q &= m_s\lambda_s \\ &= m_f c_\rho(\theta_b - \theta_f) + m_v\lambda_v \end{aligned} \qquad 13.4$$

where $c\,(\mathrm{J\,kg^{-1}{}^oC^{-1}})$ = specific heat capacity of feed liquor, $\lambda_s\,(\mathrm{J\,kg^{-1}})$ = latent heat of condensing steam, $\lambda_v\,(\mathrm{J\,kg^{-1}})$ = latent heat of vapourisation of water (Table 13.2). That is:

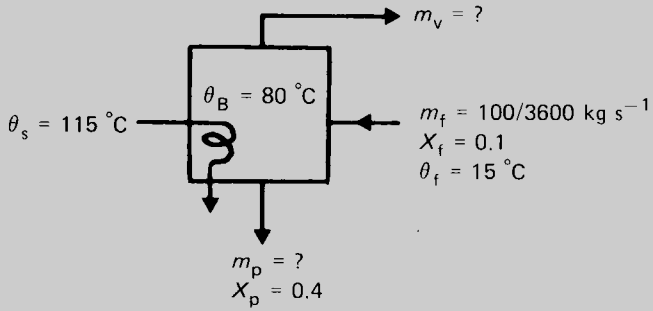
Heat supplied by steam = Sensible heat + Latent heat of vaporisation

The rate of heat transfer across evaporator walls and boundary films (Chapter 1) is found using equation (1.22). ($Q = UA(\theta_s - \theta_b)$.) For the majority of an evaporation process, the rate of heat transfer is the controlling factor and the rate of mass transfer only becomes important when the liquor becomes highly concentrated.

Sample problem 13.1

A single-effect, vertical short-tube evaporator (Section 13.1.2) is to be used to concentrate syrup from 10% solids to 40% solids at a rate of $100\,\mathrm{kg\,h^{-1}}$. The feed enters at 15ºC and is evaporated under a reduced pressure of 47.4 kPa (at 80ºC). Steam is supplied at 169 kPa (115ºC). Assuming that the boiling point remains constant and that there are no heat losses, calculate the quantity of steam used per hour and the number of tubes required. (Additional data: the specific heat of syrup is constant at $3.960\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$, the specific heat of water is $4.186\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$, the latent heat of vaporisation of the syrup is $2309\,\mathrm{kJ\,kg^{-1}}$, the overall heat transfer coefficient is $2600\,\mathrm{W\,m^{-2}\,K^{-1}}$ and the latent heat of steam is $2217\,\mathrm{kJ\,kg^{-1}}$ at 115ºC.)

Solution to Sample problem 13.1



To find the quantity of steam used per hour, we find that, from equation (13.2),

$$\frac{100}{3600} \times 0.1 = m_p \times 0.4$$

$$m_p = 0.0069\,\mathrm{kg\,s^{-1}}$$

From equation (13.1),

$$\frac{100}{3600}(1-0.1) = 0.0069\,(1-0.4) + m_v$$

$$m_v = 0.0209\,\text{kg s}^{-1}$$

From a mass balance, we find the following table.

	mass (kg s^{-1})		
	Solids	Liquid	Total
Feed	0.00278	0.025	0.0278
Product	0.00276	0.00414	0.0069
Vapour			0.0209

From equation (13.4), the heat required for evaporation is

$$Q = 0.0278 \times 3960\,(80-15) + 0.0209 \times 2309 \times 10^3$$
$$= 5.54 \times 10^4\,\text{J s}^{-1}$$

Now $\begin{matrix}\text{heat supplied by 1 kg}\\ \text{of steam per second}\end{matrix}$ = latent heat + $\begin{matrix}\text{sensible heat}\\ \text{on cooling at 80°C}\end{matrix}$

$$= 2217 \times 10^3 + 1 \times 4186 \times (115-80)$$
$$= 2.36 \times 10^6\,\text{J s}^{-1}$$

On the assumption of a heat balance in which the heat supplied by the steam equals the heat required for evaporation,

$$\text{mass of steam} = \frac{5.54 \times 10^4}{2.36 \times 10^6}$$
$$= 0.023\,\text{kg s}^{-1}$$
$$= 84.5\,\text{kg h}^{-1}$$

To find the number of tubes, we have from equation (1.22), ($Q = UA\Delta t$) that

$$5.54 \times 10^4 = 2600 \times A\,(115-80)$$

Therefore

$$A = 0.61\,\text{m}^2$$

Now

$$\text{area of one tube} = 0.025 \times 1.55 \times 3.142$$
$$= 0.122\,\text{m}^2$$

Thus,

$$\text{number of tubes} = \frac{0.61}{0.122}$$
$$= 5$$

Table 13.2 Latent heat of vaporisation of water

Temperature (°C)	Latent heat ($J\,kg^{-1}$)
0	2.494×10^6
20	2.448×10^6
40	2.402×10^6
60	2.357×10^6
80	2.309×10^6
100	2.258×10^6

Sample problem 13.2
Milk containing 3.7% fat and 12.8% total solids is to be evaporated to a produce a product containing 7.9% fat. What is the yield of product from 100 kg of milk and what is the total solids concentration in the final product, assuming that there are no losses during the process?

Solution to Sample problem 13.2

$$\text{Mass of fat in 100 kg of milk} = 100 \times 0.037$$

If Y = yield of product:

$$\text{Mass of fat in the evaporated milk} = Y \times 0.079$$

As no fat is gained or lost during the process:

$$0.79 \times Y = 3.7$$

$$\text{Yield}(Y) = 46.8 \text{ kg}$$

$$\text{Mass of solids in the milk} = 100 \times 0.128$$

If Z = % total solids in the evaporated milk

$$\text{Solids in the product} = 46.8 \times (Z/100)$$

i.e.

$$0.4684 \times Z = 12.8$$

$$Z = 27.3\%$$

Related problems are given in Chapter 1 (Sample problems 1.2, 1.7 and 1.8).

Factors influencing the rate of heat transfer
The following factors influence the rate of heat transfer and hence determine processing times and the quality of concentrated products:

- *Temperature difference between the steam and boiling liquid*. There are two options to increase the temperature difference: to increase the pressure and temperature of the steam (Chapter 12, Table 12.6) or to reduce the temperature of the boiling liquid by evaporating under a partial vacuum. In commercial vacuum evaporators the boiling

point may be reduced to as low as 40ºC. However, both methods increase the capital cost of equipment because of the extra strength required, and also the cost of energy needed for processing. The temperature difference becomes smaller as foods become more concentrated, owing to elevation of the boiling point, and the rate of heat transfer therefore falls as evaporation proceeds. In large evaporators, the boiling point of liquid at the base may be slightly raised as a result of increased pressure from the weight of liquid above (the hydrostatic head). In such cases measurement of the boiling point for processing calculations is made half-way up the evaporator.

- *Deposits on heat transfer surfaces*. The 'fouling' of evaporator surfaces reduces the rate of heat transfer. It depends on the temperature difference between the food and the heated surface and the viscosity and chemical composition of the food. For example, denaturation of proteins or deposition of polysaccharides cause the food to burn onto hot surfaces. Fouling is reduced in some types of equipment by continuously removing food from the evaporator walls (Section 13.1.2). Metal corrosion on the steam side of evaporation equipment would also reduce the rate of heat transfer, but it is reduced by anti-corrosion chemicals or surfaces. Both types of deposit are described in detail by Pulido (1984).
- *Boundary films*. A film of stationary liquid at the evaporator wall is often the main resistance to heat transfer. The thickness of the boundary film is reduced by promoting convection currents within the food or by mechanically induced turbulence (Section 13.1.2). The viscosity of many foods increases as concentration proceeds. This reduces the Reynolds number and hence reduces the rate of heat transfer (details are given in Chapter 1, Section 1.3). In addition, more viscous foods are in contact with hot surfaces for longer periods and, as a result, suffer greater heat damage.

Factors influencing the economics of evaporation

The main factors that influence the economics of evaporation are loss of concentrate or product quality (Section 13.2) and high energy consumption. Product losses are caused by *foaming*, due to proteins and carbohydrates in the food, which causes inefficient separation of vapour and concentrate, and *entrainment*, in which a fine mist of concentrate is produced during the violent boiling, and is carried out of the evaporator by the vapour. Most designs of equipment include disengagement spaces or separators to minimise entrainment.

A substantial amount of energy is needed to remove water from foods by boiling (2257 kJ per kilogram of water evaporated at 100ºC). The economics of evaporation are therefore substantially improved by attention to the design and operation of equipment and careful planning of energy use. Smith (1997) describes an energy management system used in a sugar refinery that has resulted in substantial savings in energy consumption.

Energy can be saved by re-using heat contained in vapours produced from the boiling food by:

- *vapour recompression*, in which the pressure (and therefore the temperature) of vapour is increased, using a mechanical compressor or a Venturi-type steam jet. The resulting high pressure steam is re-used as a heating medium.
- *preheating*, in which vapour is used to heat the incoming feed liquor or condensed vapour is used to raise steam in a boiler.
- *multiple effect evaporation*, in which several evaporators (or 'effects') are connected together. Vapour from one effect is used directly as the heating medium in the next.

Table 13.3 Steam consumption with vapour recompression and multiple effect evaporation

Number of effects	Steam consumption (kg per kg of water evaporated)	
	Without vapour recompression	With vapour recompression
1	1.1	0.6
2	0.6	0.4
3	0.4	0.3

From Mannheim and Passy (1974).

However, the vapour can only be used to boil liquids at a lower boiling temperature. The effects must therefore have progressively lower pressures in order to maintain the temperature difference between the feed and the heating medium.

The number of effects used in a multiple effect system is determined by the savings in energy consumption (Table 13.3) compared with the higher capital investment required, and the provision of increasingly higher vacua in successive effects (Rumsey *et al.*, 1984). In the majority of applications, three to six effects are used but up to nine effects have been reported (Anon., 1981).

Different arrangements of multiple effect evaporators are shown in Fig. 13.2 using triple-effect evaporation as an example and the relative advantages and limitations of each arrangement are described in Table 13.4.

Table 13.4 Advantages and limitations of various methods of multiple effect evaporation

Arrangement of effects	Advantages	Limitations
Forward feed	Least expensive, simple to operate, no feed pumps required between effects, lower temperatures with subsequent effects and therefore less risk of heat damage to more viscous product	Reduced heat transfer rate as the feed becomes more viscous, rate of evaporation falls with each effect, best quality steam used on initial feed which is easiest to evaporate. Feed must be introduced at boiling point to prevent loss of economy (if steam supplies sensible heat, less vapour is available for subsequent effects)
Reverse feed	No feed pump initially, best-quality steam used on the most difficult material to concentrate, better economy and heat transfer rate as effects are not subject to variation in feed temperature and feed meets hotter surfaces as it becomes more concentrated thus partly offsetting increase in viscosity	Interstage pumps necessary, higher risk of heat damage to viscous products as liquor moves more slowly over hotter surfaces, risk of fouling
Mixed feed	Simplicity of forward feed and economy of backward feed, useful for very viscous foods	More complex and expensive
Parallel	For crystal production, allows greater control over crystallisation and prevents the need to pump crystal slurries	Most complex and expensive of the arrangements, extraction pumps required for each effect

Adapted from Brennan *et al.* (1990).

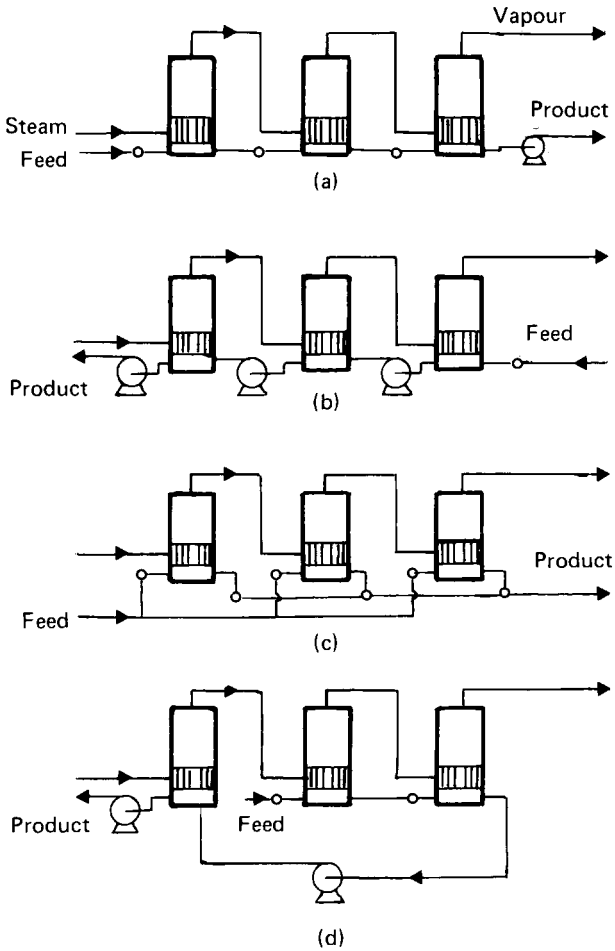


Fig. 13.2 Arrangement of effects in multiple effect evaporation: (a) forward; (b) reverse; (c) parallel; (d) mixed.
(After Brennan *et al.* (1990).)

13.1.2 Equipment

Evaporators consist of:

- a heat exchanger (termed a *calandria*) which transfers heat from steam to the food
- a means of separating the vapours produced
- a mechanical or steam ejector vacuum pump. Mechanical pumps have lower operating costs but higher capital costs than steam ejector pumps.

Separation in an evaporator is achieved by exploiting the difference in volatility between water and solutes. Ideally it should selectively remove water without changing the solute composition, so that the original product is obtained on dilution. This is approached in some equipment but, the closer to the ideal that is achieved, the higher the cost. As with other unit operations the selection of a particular method is therefore a compromise between the cost of production and the quality required in the product. The selection of an evaporator should include the following considerations:

- operating capacity (as kilograms of water removed per hour)
- degree of concentration required (as percentage of dry solids in the product)
- heat sensitivity of the product in relation to the residence time and temperature of evaporation
- the requirement for facilities to recover volatiles
- ease of cleaning
- reliability and simplicity of operation
- size of the evaporator in relation to its capacity
- capital and operating costs in relation to capacity and product quality (Mannheim and Passy, 1974).

Energy consumption is a major factor in evaporation, and significant developments have taken place in computer control of evaporators (for example Anon. (1986), also Chapter 2). The most widespread evaporators for juice concentration are falling film evaporators (see below) and an example of these is the 'thermally accelerated short-time evaporator' (TASTE) (Kennedy *et al.*, 1983). Under manual operation, the steam flow is set at a constant rate and the juice flow is manually adjusted to obtain the required concentration in the final product. This may fluctuate by a several ^{o}Brix due to changes in feed composition or other variables in the process and requires skill and experience by the operator to correct the deviation from the required concentration. This results in both off-specification products and wastage of energy. Chen *et al.* (1981) describe computer control systems, involving control of steam and juice temperatures and flowrates to improve the economy of TASTE evaporators.

In some applications it may be more cost effective to combine two types of evaporator, for example initial concentration of the bulk liquor in a low cost evaporator followed by final concentration of the smaller volume of heat sensitive liquor in a more expensive, but less damaging evaporator as the second effect. The majority of evaporator designs operate continuously but batch boiling pans are used for the preparation of small quantities of materials, or in applications where flexibility is required for frequent changes of product.

Natural circulation evaporators

Open- or closed-pan evaporators are hemispherical pans, heated directly by gas or electrical resistance wires or heated indirectly by steam passed through internal tubes or an external jacket. For vacuum operation, they are fitted with a lid. A stirrer or paddle is used to increase the rate of heat transfer and to prevent food from burning onto the pan (see Chapter 5, Fig. 5.8). They are similar in appearance to jacketed mixing vessels. They have relatively low rates of heat transfer (Table 13.5) and low energy efficiencies, and they cause damage to heat-sensitive foods. However, they have low capital costs, are relatively easy to construct and maintain and are flexible for applications where frequent changes of product are likely, or when used for relatively low or variable production rates. They have therefore found wide application in the preparation of ingredients such as sauces and gravies or in the manufacture of jam and other preserves (see, for example, Darrington, 1982).

The *short-tube evaporator* is an example of a tube-and-shell heat exchanger, also used in pasteurisation (Chapter 11) and heat sterilisation (Chapter 12). It consists of a vessel (or shell) which contains a vertical, or less commonly horizontal bundle of tubes. The vertical arrangement of tubes promotes natural convection currents and therefore higher rates of heat transfer (Fig. 13.3). Feed liquor is heated by steam condensing on the outside

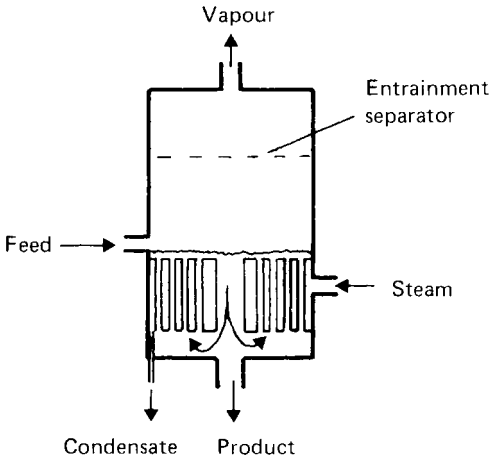


Fig. 13.3 Vertical short-tube evaporator. (After Karel (1975).)

of the tubes and rises through the tubes, boils and recirculates through a central downcomer tube. *External calandria evaporators* are tube-and-shell heat exchangers which are fitted with an external pipe for recirculation of the product. This increases convection currents and rates of heat transfer, and the calandria is easily accessible for cleaning. They are suitable for concentrating heat-sensitive foods, including dairy products and meat extracts, when operated under partial vacuum.

These evaporators have low construction and maintenance costs, high flexibility and higher rates of heat transfer than open or closed pans, when used with relatively low-viscosity liquids (Table 13.5). They are generally unsuited to high-viscosity liquors as there is poor circulation of liquor and a high risk of food burning onto the tube walls. They are used for concentrating syrups, salt and fruit juices.

Long-tube evaporators consist of a vertical bundle of tubes, each up to 5 cm in diameter, contained within a steam shell 3–15 m high. Liquor is heated almost to boiling point before entering the evaporator. It is then further heated inside the tubes and boiling commences. The expansion of steam forces a thin film of rapidly concentrating liquor up the walls of each tube (Fig. 13.4). The concentrate is separated from the vapour and removed from the evaporator, passed to subsequent effects in a multiple-effect system, or recirculated. Vapour is re-used in multiple-effect or vapour recompression systems (Section 13.1.1).

For low-viscosity foods (for example milk), the thin film of liquor is forced up the evaporator tubes and this arrangement is therefore known as a *climbing-film evaporator*. For more viscous foods, or those that are very heat sensitive, (for example yeast extracts, fruit juices (Anon., 1981) and in starch processing), the feed is introduced at the top of the tube bundle in a *falling-film evaporator*. The force of gravity supplements the forces arising from expansion of the steam, to produce very high liquor flow rates (up to 200 m s^{-1} at the end of 12 m tubes) and short residence times (typically 5–30 s). Multiple-effect systems, capable of evaporating 45 000 l of milk per hour, have been described (Anon., 1986). Both types of long-tube evaporator are characterised by high heat transfer coefficients (Table 13.5) and efficient energy use (0.3–0.4 kg of steam per kilogram of water evaporated in multiple-effect systems). Falling film evaporators are now the most commonly used in the food industry and are described in detail by Burkart and Wiegand (1987).

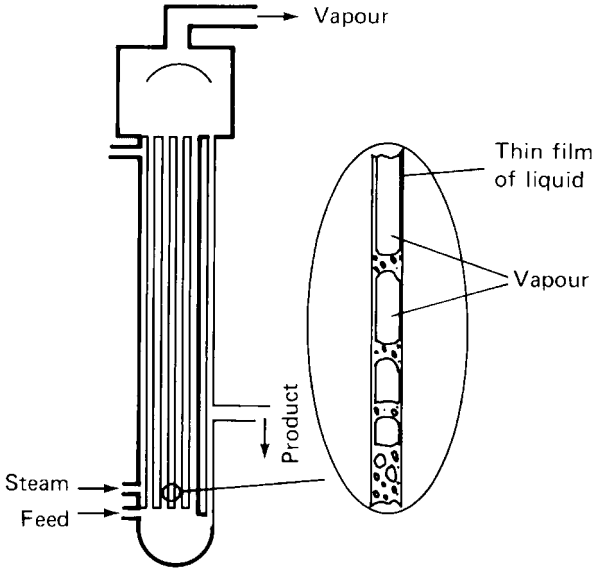


Fig. 13.4 Climbing-film evaporator.

Forced circulation evaporators

In forced-circulation evaporators a pump or scraper assembly moves the liquor, usually in thin layers, and thus maintains high heat transfer rates and short residence times (Table 13.5). This also results in more compact equipment and higher production rates but increases both the capital and the operating costs of the equipment.

Plate evaporators are similar in construction to the heat exchangers used for pasteurisation and ultra high-temperature (UHT) sterilisation (Chapters 11 and 12). However, in this case the climbing- and falling-film principle is used to concentrate liquids in the spaces between plates. The number of climbing- or falling-film sections fitted within a single machine depends on the production rate and degree of concentration required. The mixture of vapour and concentrate is separated outside the evaporator. Despite the high capital investment, these types of evaporator have high rates of heat transfer, short residence times and high energy efficiencies (Table 13.5). They are compact, capable of high throughputs and easily dismantled for maintenance and inspection. They are more suitable for heat-sensitive foods of higher viscosity (0.3–0.4 $N\,s\,m^{-2}$) including yeast extracts, dairy products, fruit juices, low alcohol drinks and meat extracts. Their advantages, compared to other falling film evaporators, are described by Olsson (1988).

The *expanding-flow evaporator* uses similar principles to the plate evaporator but has a stack of inverted cones instead of a series of plates. Feed liquor flows to alternate spaces between the cones from a central shaft and evaporates as it passes up through channels of increasing flow area (hence the name of the equipment). Steam is fed down alternate channels. The vapour-concentrate mixture leaves the cone assembly tangentially and is separated by a special design of shell which induces a cyclone effect. This evaporator has a number of advantages including compactness, short residence times and a high degree of flexibility achieved by changing the number of cones.

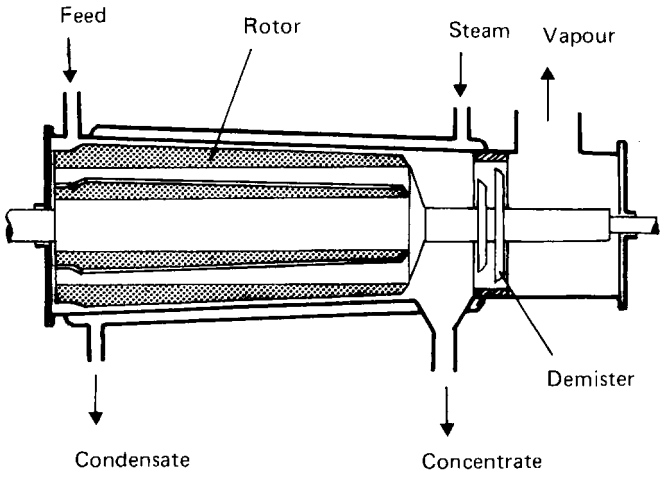


Fig. 13.5 Mechanical thin film evaporator.
(After Leniger and Beverloo (1975).)

Mechanical (or agitated) thin-film evaporators

Scraped- or wiped-surface evaporators are characterised by differences in the thickness of the film of food being processed. Wiped-film evaporators have a film thickness of approximately 0.25 mm whereas in scraped-film evaporators it is up to 1.25 mm. Both types consist of a steam jacket surrounding a high-speed rotor, fitted with short blades along its length (Fig. 13.5). The design is similar to a scraped-surface aseptic steriliser (Chapter 12). Feed liquor is introduced between the rotor and the heated surface and evaporation takes place rapidly as a thin film of liquor is swept through the machine by the rotor blades. The blades keep the film violently agitated and thus promote high rates of heat transfer and prevent the product from burning onto the hot surface (Anon., 1981). The residence time of the liquor is adjusted between 0.5 s and 100 s depending on the type of food and the degree of concentration required.

This type of equipment is highly suited to viscous (up to $20\,N\,s\,m^{-2}$) heat-sensitive foods or to those that are liable to foam or foul evaporator surfaces (for example fruit pulps and juices, tomato paste, meat extracts, honey, cocoa mass, coffee and dairy products). However, the capital costs are high owing to the precise alignment required between the rotor and wall. Operating costs are also high as only single effects are possible, which reduces the throughput and gives poor steam economy. It is therefore used for 'finishing' highly viscous products after concentration in other equipment where there is less water to be removed, the product is valuable and there is a substantial risk of heat damage.

A second design of mechanical thin-film evaporator is the *Centri-therm evaporator* which, although similar in appearance to the expanding flow evaporator, operates using a different principle. Here, liquor is fed from a central pipe to the undersides of rotating hollow cones. It immediately spreads out to form a layer approximately 0.1 mm thick. Steam condenses on the other side of each cone, and rapidly evaporates the liquor. In contrast with the expanding-flow evaporator, in which liquid is moved by vapour pressure, the Centri-therm employs centrifugal force to move the liquor rapidly across the heated surface of the cone. Residence times are 0.6–1.6 s (Lewicki and Kowalczyk,

Table 13.5 Comparison of residence times and heat transfer coefficients in selected evaporators

Type of evaporator	Number of stages	Residence time (approximate)	OHTC ($W\,m^{-2}\,K^{-1}$) Low viscosity	OHTC ($W\,m^{-2}\,K^{-1}$) High viscosity
Open or vacuum pan	Single	30 min to several hours	500–1000	< 500
Vertical short tube	Single	–	570–2800	–
Climbing film	Single	10–60 sec	2250–6000	< 300
Falling film	Single	5–30 sec	2000–3000	–
Plate	Three	2–30 sec	2000–3000	–
Expanding flow	Two	0.5–30 sec	2500	–
Agitated film	Single	20–30 sec	2000–3000	1700
Centri-Therm	Single	1–10 sec	8000	–

Adapted from Mannheim and Passy (1974) and Earle (1983).

1980), even with concentrated liquors (up to $20\,N\,s\,m^{-2}$). Very high heat transfer coefficients and short residence times are possible (Table 13.5). This is due in part to the thin layers of liquor but also to the droplets of condensed steam which are flung from the rotating cones as fast as they are formed. There is therefore no boundary film of condensate to impede heat transfer. The equipment produces a concentrate which, when rediluted, has sensory and nutritional qualities that are virtually unchanged from those of the feed material. It is used for coffee and tea extracts, meat extract, fruit juices (Fischer *et al.*, 1983) and enzymes for use in food processing.

13.2 Effect on foods

Aroma compounds that are more volatile than water are thus lost during evaporation. This reduces the sensory characteristics of most concentrates; in fruit juices this results in a loss of flavour, although in some foods the loss of unpleasant volatiles improves the product quality (for example in cocoa (Anon., 1981) and milk). Some volatiles can be recovered and retained in the product:

- by volatile recovery by vapour condensation and fractional distillation
- by stripping volatiles from the feed liquor with inert gas and adding them back after evaporation (see also the spinning cone column in Section 13.3).

These methods are described in more detail in Section 13.3. The advantages and limitations of each procedure have been reviewed by Mannheim and Passy (1972) and Thijssen (1970). Flash coolers, in which the food is sprayed into a vacuum chamber, are used to cool a viscous product rapidly and hence to reduce heat damage.

Evaporation darkens the colour of foods, partly because of the increase in concentration of solids, but also because the reduction in water activity promotes chemical changes, (for example Maillard browning (Chapter 1)). As these changes are time and temperature dependent, short residence times and low boiling temperatures produce concentrates which have a good retention of sensory and nutritional qualities. A comparison of nutrient losses in milk preserved by evaporation and UHT sterilisation is shown in Table 13.6. Vitamins A and D and niacin are unaffected. Additional vitamin losses occur during storage (for example 50% loss of vitamin C in marmalade over 12 months at 18ºC (Lincoln and McCay, 1945) and 10% loss of thiamin over 24 months in peanut butter at 18ºC).

Table 13.6 Vitamin losses in concentrated and UHT sterilised milk

Product	Loss (%)				
	Thiamin	Vitamin B_6	Vitamin B_{12}	Folic acid	Ascorbic acid
Evaporated milk	20	40	80	25	60
Sweetened condensed milk	10	< 10	30	25	25
UHT sterilised milk	< 10	< 10	< 10	< 10	< 25

From Porter and Thompson (1976).

13.3 Distillation

Although common in the chemical industry, distillation in food processing is mostly confined to the production of alcoholic spirits and separation of volatile flavour and aroma compounds (for example, production of essential oils by steam distillation).

When a food that contains components having different degrees of volatility is heated, those that have a higher vapour pressure (more volatile components) are separated first. These are termed the 'distillate' and components that have a lower volatility are termed 'bottoms' or residues. Although batch distillation (in 'pot stills') remains in use in some whisky and other spirit distilleries, most industrial distillation operations use more economical continuous distillation columns (Fig. 13.6a,b) (Kent and Evers, 1994). Feed liquor flows continuously through the column and as it is heated, volatiles are produced and separated at the top of the column as distillate and the residue is separated at the base. In order to enhance the separation of these components and equilibrium conditions between the liquid and vapour phases, a proportion of the distillate is added back to the top of the column (reflux) and a portion of the bottoms is vapourised in a reboiler and added to the bottom of the column. Columns are filled with either a packing material (typically ceramic, plastic or metal rings) or fitted with perforated trays, both of which increase the contact between liquid and vapour phases.

A more recent development is the use of a 'spinning cone column' to remove volatile components from liquids. It is used to recover flavours from beer, coffee, tomato products and fruit juices, to produce low-alcohol wines and beers and to remove off-flavours. The equipment consists of a column containing a series of rotating inverted cones, which are intermeshed with stationary cones attached to the column wall. Steam or nitrogen is supplied to the base of the column and the feed liquor enters at the top. Thin turbulent films are produced over the large surface area of the cones and rapid separation takes place. The gas passes out of the top of the column and volatile aroma compound are condensed and collected. Because separation is achieved by mechanical energy, rather than heat, there is less damage to flavours and lower energy consumption. The equipment is also considerably smaller than a packed column having an equivalent throughput (Schofield, 1995).

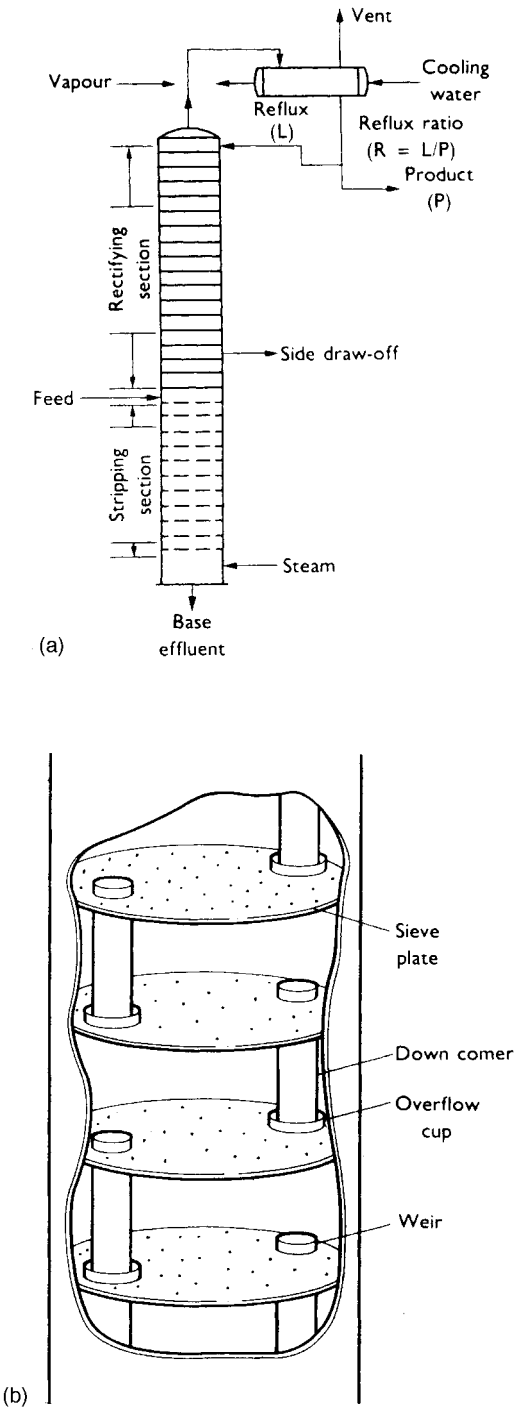


Fig. 13.6 (a) Continuous distillation column and (b) internal plates in the column to promote cross-flow.
(From Panek and Boucher (1989).)

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